

1. 解, a) $\min W = 5y_1 + 8y_2 + 20y_3$ 对偶: $\max Z = x_1 + 2x_2 + 3x_3 + 4x_4$

$$-y_1 + 6y_2 + 12y_3 \geq 1$$

$$y_1 + 7y_2 - 9y_3 \geq 2$$

$$-y_1 + 3y_2 - 9y_3 \leq 3$$

$$-3y_1 - 5y_2 + 9y_3 = 4$$

$$y_1 \text{ 无约束}, y_2 \leq 0, y_3 \geq 0$$

$$-x_1 + x_2 - x_3 - 3x_4 = 5$$

$$6x_1 + 7x_2 + 3x_3 - 5x_4 \geq 8$$

$$12x_1 - 9x_2 - 9x_3 + 9x_4 \leq 20$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \leq 0, x_4 \text{ 无约束}$$

b) $\max W = \sum_{i=1}^m a_i y_i + \sum_{j=1}^n b_j y_{m+j}$ 对偶 $\min Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$

$$y_i + y_{m+j} \leq c_{ij} \quad \begin{matrix} i=1, 2, \dots, m \\ j=1, 2, \dots, n \end{matrix}$$

$$y_i, y_{m+j} \text{ 无约束}$$

$$\sum_{j=1}^n x_{ij} = a_i \quad (i=1, \dots, m)$$

$$\sum_{i=1}^m x_{ij} = b_j \quad (j=1, \dots, n)$$

$$x_{ij} \geq 0$$

2. 解: (4) $\begin{pmatrix} 2x_1 + x_3 + x_4 - 8 \\ 2x_1 + 2x_2 + x_3 + 2x_4 - 12 \end{pmatrix} = 0$

$$(x_1, x_2, x_3, x_4) \begin{pmatrix} 8 \\ 12 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$\therefore \begin{cases} 2x_1 + x_3 + x_4 = 8 \\ 2x_1 + 2x_2 + x_3 + 2x_4 = 12 \\ x_1 = 0 \\ x_2 = 0 \end{cases}$$

$$\therefore \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 4 \\ x_4 = 4 \end{cases}$$

3. 解:

$$\begin{array}{cccccc} & & & 1 & 1 & 0 & 0 \\ x_3 & 0 & -4 & -2 & -1 & 1 & 0 \\ x_4 & 0 & -7 & -1 & -7 & 0 & 1 \\ & & & 1 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{cccccc} & & & 1 & 1 & 0 & 0 \\ x_1 & 1 & 2 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ x_4 & 0 & -5 & 0 & -\frac{13}{2} & -\frac{1}{2} & 1 \\ & & & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{array}$$

$$\begin{array}{cccccc} & & & 1 & 1 & 0 & 0 \\ x_1 & 1 & \frac{21}{13} & 1 & 0 & -\frac{7}{13} & \frac{1}{13} \\ x_2 & 1 & \frac{10}{13} & 0 & 1 & \frac{1}{13} & -\frac{7}{13} \\ & & & 0 & 0 & \frac{6}{13} & \frac{1}{13} \end{array}$$

$$\therefore Z_{\min} = \frac{21}{13} + \frac{10}{13} = \frac{31}{13}$$